

Representation as a Computational Resource in Quantum Compilation

Approximate recoverability tradeoffs and model-specific fault-tolerant consequences

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Group seminar — AQIS 2026 poster session rehearsal

One-slide thesis

一句话论点

Fault-tolerant compilation **deliberately destroys structure.**

Putting it back together has a price, payable in **exactly two currencies:**

bits of compiler state, or magic states.

Design rule with a proof: **randomize and lower late.**

容错编译会主动破坏结构。把结构装回去是有价钱的，而且只有两种货币：编译器状态的比特，或者魔术态。设计规则：randomize and lower late（晚随机化、晚lowering）。

► Terminology and toolkit 术语与工具字典

Packing: why distinguishable settings are countable 为什么“可区分的设置”只有可数个

Calibration: K_ϵ was not chosen by me 字母表大小 K_ϵ 不是我挑的

Dispersal: no record leaks the aggregate 没有任何一条记录泄露 aggregate

The lower bounds: one argument, three variations 下界：一个论证的三次变奏

Upper bounds: a frontier, not a vacuous floor 上界：这是前沿，不是空洞的下界

Qubits, unitaries, and Z-type characters

比特、酉算子与 Z 型特征标

State space $\mathbb{C}^{2^n}$, computational basis $|z\rangle, z \in \mathbb{F}_2^n$.

For $a \in \mathbb{F}_2^n$: $Z^a := Z^{a_1} \otimes \cdots \otimes Z^{a_n}$ is **diagonal**, and

$$Z^a |z\rangle = (-1)^{a \cdot z} |z\rangle =: \chi_a(z) |z\rangle,$$

$$a \cdot z = \sum_i a_i z_i \bmod 2.$$

Every gate is a $2^n \times 2^n$ unitary; a circuit is a product of unitaries.

$\chi_a : \mathbb{F}_2^n \rightarrow \{\pm 1\}$ is the **character** of a : a homomorphism from the additive group $\mathbb{F}_2^n$ to $\{\pm 1\}$.

This is the only “group theory” in the talk. “ a_j linearly independent” will become “the characters jointly realize every sign pattern.”

全场只用到这一点群论。之后说“ a_j 线性无关”，意思就是“这些 character 联合起来能取遍每一种符号模式”。

Rotations, and why phase layers aggregate

旋转，以及相位层为什么能聚合

For Hermitian P with $P^2 = I$:

$$R_P(\theta) := e^{-i\theta P/2} = \cos(\theta/2) I - i \sin(\theta/2) P.$$

Derivation: expand the exponential series and use $P^2 = I$ to collapse the even terms into cos and the odd terms into sin.

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All Z^a are diagonal $\Rightarrow$ pairwise commuting $\Rightarrow$

$$\prod_j R_{P_j}(\theta_j) = \exp\left(-\frac{i}{2} \sum_j \theta_j P_j\right).$$

所有 Z^a 都是对角的，因此两两对易，于是一层相位旋转的乘积就等于把指数相加。

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► Must answer

This is the physical premise of the whole paper. Commutation is what makes “add the coefficients across rounds” a legal simplification. Insert a non-commuting layer (the R_X layers of a generic Trotter step) and the exponents no longer add — **the aggregate does not exist at all.**

对易性是全文的物理前提——它才让”跨轮次把系数相加”成为合法化简。插进一个不对易的层，aggregate 根本就不存在。

Clifford cheap, T expensive: the economics

Clifford 便宜、 T 昂贵：成本结构

Clifford group = the *normalizer* of the Pauli group: C is Clifford iff C^+PC is again a Pauli (up to sign) for every Pauli P . Examples: H, S, CNOT . Clifford gates only move Paulis to Paulis, so their effect on Pauli errors is trackable with classical bits.

Object	Meaning	Cost
logical qubit	surface code, $O(d^2)$ physical qubits, code distance d	hundreds-thousands of physical qubits
Clifford gate	implementable fault-tolerantly by the code itself	$\approx$ free
$T = \text{diag}(1, e^{i\pi/4})$	needs a magic state $ T\rangle$	must be distilled in a dedicated factory
T -count	—	the unit of account of FT computation

只需接受一件事：Clifford 门几乎免费， T 门是唯一真正花钱的门——它要在专门的 factory 里蒸馏 magic state。所以容错计算的记账单位是 T -count。

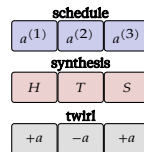
Three mechanisms that disperse an angle

把一个角度打散的三种机制

- ① **Scheduling.** Time-dependent / variational layers split one accumulated phase across r rounds.
- ② **Synthesis (Ross-Selinger).** An angle becomes a Clifford+ T word in which no single letter carries the angle. Typical T -count = $3 \log_2(1/\epsilon) + O(\log \log(1/\epsilon))$.
- ③ **Randomized compiling (twirling).** Each rotation is multiplied by a random sign $s \in \{\pm 1\}$ with Pauli compensation, so coherent noise averages into Pauli noise.

Two presentations:

- *folded*: sign absorbed at emission, $a \mapsto sa \bmod Q$;
- *tracked*: signs live in a classical *Pauli frame* register carried by the control stack.



三种机制都会把一个角度打散：schedule 把它分到 r 轮，synthesis 把它变成一串 Clifford+ T 字母，twirling 再给每个旋转乘上随机符号。

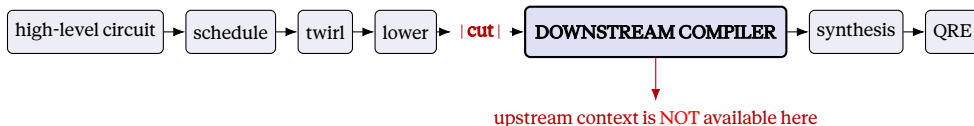
► Must answer

The bound is carried by the **schedule alone**. Sign normalization is a bijection of the residue alphabet, so the standard **folded (compile-time)** presentation inherits the bound verbatim. **Nothing depends on tracking Pauli frames.**

界只由 schedule 承担。符号归一化是残数字母表上的双射，所以标准的 folded（编译期）写法逐字继承这个界，完全不依赖追踪 Pauli frame。

Where the compiler sits: downstream

编译器的位置：在下游



Three real instances:

- a cloud transpiler receiving a flat QASM/QIR program without the user's high-level intent;
- a real-time stream optimizer inside the classical control stack;
- a vendor-side pass post-processing an already emitted instruction stream.

If the compiler can see the circuit before randomization and lowering, the theorems do not apply — and that boundary is itself the content of the design rule.

若编译器能在 randomization 和 lowering 之前看到电路，定理就不适用——而这条边界本身就是设计规则的内容。

Also worth knowing (rapid fire)

顺带一提（快速扫过）

- **fidelity** — a closeness measure for states/operations; *not used here*, we use operator-norm distances because they satisfy the triangle inequality (needed for packing arguments).
- **oracle** — a black-box subroutine charged per call; appears only in scope statements (“an uncharged random-access input oracle would be a different model”).
- **FPGA / control stack** — the classical electronics adjacent to the chip must resolve corrections *before the next non-Clifford operation*, under hard latency and buffer budgets. **This is why the bounded-memory streaming model is an engineering constraint, not a mathematical convenience.**
- **spacetime volume** — physical qubits $\times$ time (qubit-seconds).
- **QRE** — quantum resource estimator: logical circuit + architecture assumptions $\mapsto$ T count, physical qubits, cycles, factories.
- **lowering / IR / transpiler** — lowering produces a *flat instruction stream*; the graph/table the compiler keeps internally is the *IR*.
- **QAOA cost layer** — $e^{-i\gamma C}$, $C = \sum_j w_j Z_{u_j} Z_{v_j}$: one R_{ZZ} per edge.

这一页只需记住两点：我们用算子范数而不是 fidelity（因为 packing 论证需要三角不等式）；有界内存的流式模型来自控制栈真实的延迟与缓冲预算，不是数学上的方便假设。

Mathematical toolkit (1/2): circle distance

数学工具 (1/2): 圆周距离

$$\langle\langle u \rangle\rangle := \min_{k \in \mathbb{Z}} |u - 2\pi k| \in [0, \pi], \quad d_{\circ}(\alpha, \beta) := \langle\langle \alpha - \beta \rangle\rangle.$$

$d_{\circ}$ is a metric on $\mathbb{R}/2\pi\mathbb{Z}$ (the triangle inequality holds). The identity used throughout:

$$|e^{iu} - 1| = 2 \sin(\langle\langle u \rangle\rangle/2). \quad (0.1)$$

$\langle\langle \cdot \rangle\rangle$ 是圆周距离，已经把”绕过 2π ”吸收进定义。(0.1) 把弦长换算成角度，是全场引用最多的式子之一。

Verification. $|e^{iu} - 1| = 2|\sin(u/2)|$; for $u \in [0, \pi]$, $\langle\langle u \rangle\rangle = u$; for $u \in [\pi, 2\pi]$, $\langle\langle u \rangle\rangle = 2\pi - u$ and $2 \sin(\frac{2\pi-u}{2}) = 2 \sin(\pi - \frac{u}{2}) = 2 \sin \frac{u}{2}$.

Mathematical toolkit (2/2): four information-theoretic facts

数学工具 (2/2): 四个信息论事实

Prefix-free code: no codeword is a prefix of another; concatenations are uniquely parseable without out-of-band lengths. All five serialization fields of the model are such codes.

Fact K (Kraft). $\sum_i 2^{-\ell_i} \leq 1$, hence

$$N \cdot 2^{-\max_i \ell_i} \leq 1 \implies \max_i \ell_i \geq \log_2 N.$$

Proof (one line). Map codeword w to the dyadic interval $I_w = [0.w, 0.w + 2^{-|w|})$; w is a prefix of w' iff $I_{w'} \subseteq I_w$; prefix-freeness $\implies$ the I_i are pairwise disjoint $\implies \sum 2^{-\ell_i} = \sum |I_i| \leq 1$.

Fact S (Shannon). Any prefix-free code satisfies $\mathbb{E}|E(X)| \geq H(X)$.

With $|\mathcal{X}| = K^m$:

$$L_\delta(m, K) := (1 - \delta) m \log_2 K - h_2(\delta).$$

(0.2)

Kraft 给出“最长码字至少 $\log_2 N$ 比特”；Fano 给出“能以 δ 错误率猜中，就意味着互信息足够大”。(0.2) 是后面所有下界的右端。

Fact F (Fano). If $\hat{X} = f(Y)$ and $\Pr[\hat{X} \neq X] \leq \delta$, then

$$H(X | Y) \leq h_2(\delta) + \delta \log_2(|\mathcal{X}| - 1).$$

Intuition (this is the proof skeleton): given Y , to finish describing X you need only say (i) did the guess fail — $h_2(\delta)$ bits; (ii) if so, which of the other $|\mathcal{X}| - 1$ values — on average $\delta \log_2(|\mathcal{X}| - 1)$ bits.

Corollary F' (the only form used). For X uniform on $\mathcal{X}$:

$$I(X; Y) \geq (1 - \delta) \log_2 |\mathcal{X}| - h_2(\delta).$$

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The projective distance, and why

投影距离，以及为什么用它

Definition (projective distance)

$$d_{\text{proj}}(U, V) := \inf_{\varphi \in \mathbb{R}} \|U - e^{i\varphi} V\|_{\text{op}}.$$

- **Infimum attained:** $\varphi \mapsto \|U - e^{i\varphi} V\|_{\text{op}}$ is continuous on the compact circle $\{e^{i\varphi}\} \cong S^1$.
- **Why quotient the global phase:** U and $e^{i\varphi} U$ are physically indistinguishable. Without the quotient we would count physically identical objects as distinct, and **the packing count would simply be wrong.**

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Claim (reduction to the relative unitary)

$$d_{\text{proj}}(U, V) = \inf_{\varphi} \|UV^\dagger - e^{i\varphi} I\|_{\text{op}}.$$

Proof. $\|AB\|_{\text{op}} = \|A\|_{\text{op}}$ for unitary B ; take $B = V^\dagger$. ■

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The problem is now purely geometric: **given the spectrum of a unitary (a set of points on the unit circle), rotate the whole circle to sit as close to 1 as possible — how far is the worst point?**

问题现在纯粹是几何的：给定一个 unitary 的谱（单位圆上一组点），整体旋转让它尽量靠近 1——最坏的那个点还有多远？

Lemma 1.3: the two-point bound, and where the “4” comes from

引理 1.3: 两点界, 以及“4”从哪来

Lemma (two-point bound)

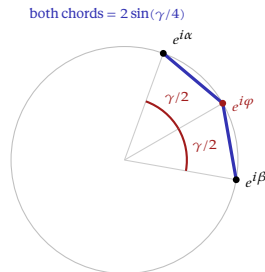
Let $\alpha, \beta \in \mathbb{R}/2\pi\mathbb{Z}$ with $d_o(\alpha, \beta) = \gamma \in [0, \pi]$. Then

$$\forall \varphi: \max(|e^{i\alpha} - e^{i\varphi}|, |e^{i\beta} - e^{i\varphi}|) \geq 2 \sin(\gamma/4),$$

with equality when φ is the midpoint of the short arc between α and β .

圆上两点相距 γ 时, 无论把全局相位转到哪里, 总有一点离它至少 $2 \sin(\gamma/4)$; 取短弧中点时取等号。

Proof. Triangle inequality on d_o : $d_o(\alpha, \varphi) + d_o(\varphi, \beta) \geq \gamma \Rightarrow \max(d_o(\alpha, \varphi), d_o(\beta, \varphi)) \geq \gamma/2$. WLOG $d_o(\alpha, \varphi) \geq \gamma/2$. By (0.1), $|e^{i\alpha} - e^{i\varphi}| = 2 \sin(d_o(\alpha, \varphi)/2)$; since $t \mapsto 2 \sin(t/2)$ increases on $[0, \pi]$ and $d_o \in [0, \pi]$, $|e^{i\alpha} - e^{i\varphi}| \geq 2 \sin((\gamma/2)/2) = 2 \sin(\gamma/4)$. Equality: the short-arc midpoint gives $d_o(\alpha, \varphi) = d_o(\beta, \varphi) = \gamma/2$, and both chords equal $2 \sin(\gamma/4)$. ■



► Why $\sin(\Theta/4)$ and not $\sin(\Theta/2)$?

Two factors of 2 compose: **(i)** the optimal global phase sits **in the middle** $\Rightarrow$ angular distance halves from γ to $\gamma/2$; **(ii)** the chord formula $|e^{iu} - 1| = 2 \sin(u/2)$ carries a $/2$ of its own. $\gamma/2$ then $/2$ gives $\gamma/4$.

两个 2 叠加: 最优全局相位落在中点, 角距离从 γ 减半到 $\gamma/2$; 弦长公式又自带一个 $/2$ 。

Lemma 1.4: the character surjectivity lemma

引理 1.4: 特征标满射引理

Lemma (character surjectivity)

For $a_1, \dots, a_m \in \mathbb{F}_2^n$:

$a_1, \dots, a_m$ linearly independent over $\mathbb{F}_2 \iff \Phi : \mathbb{F}_2^n \rightarrow \{\pm 1\}^m, z \mapsto (\chi_{a_1}(z), \dots, \chi_{a_m}(z))$ surjective.

a_j 在 $\mathbb{F}_2$ 上线性无关, 等价于这些 character 联合起来能取遍 $\{\pm 1\}^m$ 的每一种符号模式。

Proof. Let $A \in \mathbb{F}_2^{m \times n}$ have rows a_j and $L(z) = Az$. The map $\mathbb{F}_2^m \rightarrow \{\pm 1\}^m, (b_j) \mapsto ((-1)^{b_j})$, is a bijection, and $\Phi = (-1)^{L(\cdot)}$. Hence Φ surjective $\iff L$ surjective $\iff \text{rank } A = m \iff \text{rows independent}$. ■

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This is the entire group-theoretic content — and the origin of the paper's **most attackable assumption**. Lose $\mathbb{F}_2$ -independence and Φ is no longer surjective; the very first step of Theorem 1.7 below cannot be executed, and the coefficients are no longer identifiable.

这就是全部的群论内容，也是全文最容易被攻击的假设。失去 $\mathbb{F}_2$ -无关性， Φ 不再满射，Theorem 1.7 的第一步就执行不了，系数也不再可辨识。

Definitions: wrap margin and the phase family

定义：绕转裕度与相位族

Definition (wrap margin)

For alphabet $[K]_0 = \{0, \dots, K-1\}$, $K \geq 2$, step $\Delta > 0$:

$$\gamma(K, \Delta) := \min_{1 \leq d \leq K-1} \langle\langle d\Delta \rangle\rangle.$$

wrap margin: 两个系数最少能差多少角度。《·》已经把绕转算进去了。

“The least angle by which two coefficients can differ” — and $\langle\langle \cdot \rangle\rangle$ already accounts for wrapping past 2π .

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Definition (phase family)

$P_j = Z^{a_j}$ with $a_1, \dots, a_m \in \mathbb{F}_2$ -independent,

$$U_x := \exp\left(-\frac{i}{2} \sum_{j=1}^m x_j \Delta P_j\right), \quad x \in [K]_0^m.$$

相位族: 每个坐标 x_j 取自 K 元字母表, 乘上步长 Δ 作为 P_j 的转角, 共 K^m 个目标。

Theorem 1.7: exact worst-pair separation

定理 1.7: 最坏点对的精确分离

Theorem (exact separation)

For every $m \geq 1$,

$$\min_{x \neq y} d_{\text{proj}}(U_x, U_y) = 2 \sin\left(\frac{\gamma(K, \Delta)}{4}\right).$$

In particular the family is projectively injective $\iff \gamma(K, \Delta) > 0$.

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Proof, direction ($\geq$). Take $x \neq y, s := x - y \neq 0$, pick j^* with $s_{j^*} \neq 0$. By Claim 1.2 consider

$$W := U_x U_y^\dagger = \exp\left(-\frac{i\Delta}{2} \sum_j s_j P_j\right), \quad W|z\rangle = \lambda(z)|z\rangle, \quad \lambda(z) = \exp\left(-\frac{i\Delta}{2} \sum_j s_j \chi_{a_j}(z)\right).$$

By Lemma 1.4, Φ is surjective, so choose z, z' with $\chi_{a_j}(z) = \chi_{a_j}(z')$ for $j \neq j^*$, and $\chi_{a_{j^*}}(z) = +1, \chi_{a_{j^*}}(z') = -1$. Then

$$\arg \lambda(z) - \arg \lambda(z') = -\frac{\Delta}{2} (s_{j^*} - (-s_{j^*})) = -\Delta s_{j^*} \implies d_o(\arg \lambda(z), \arg \lambda(z')) = \langle \Delta s_{j^*} \rangle \geq \gamma(K, \Delta),$$

since $1 \leq |s_{j^*}| \leq K-1$. Finally $\|W - e^{i\varphi} I\|_{\text{op}} = \max_z |\lambda(z) - e^{i\varphi}| \geq \max(|\lambda(z) - e^{i\varphi}|, |\lambda(z') - e^{i\varphi}|)$, so taking $\inf_{\varphi}$ and applying Lemma 1.3 gives $d_{\text{proj}}(U_x, U_y) \geq 2 \sin(\gamma/4)$. ■

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By Lemma 1.4, Φ is surjective, so choose z, z' with $\chi_{a_j}(z) = \chi_{a_j}(z')$ for $j \neq j^*$, and $\chi_{a_{j^*}}(z) = +1, \chi_{a_{j^*}}(z') = -1$. Then

$$\arg \lambda(z) - \arg \lambda(z') = -\frac{\Delta}{2} (s_{j^*} - (-s_{j^*})) = -\Delta s_{j^*} \implies d_o(\arg \lambda(z), \arg \lambda(z')) = \langle \Delta s_{j^*} \rangle \geq \gamma(K, \Delta),$$

since $1 \leq |s_{j^*}| \leq K-1$. Finally $\|W - e^{i\varphi} I\|_{\text{op}} = \max_z |\lambda(z) - e^{i\varphi}| \geq \max(|\lambda(z) - e^{i\varphi}|, |\lambda(z') - e^{i\varphi}|)$, so taking $\inf_{\varphi}$ and applying Lemma 1.3 gives $d_{\text{proj}}(U_x, U_y) \geq 2 \sin(\gamma/4)$. $\blacksquare$

Proof, direction ($\leq$): equality is attained. Pick $d \in \{1, \dots, K-1\}$ attaining γ and take $x - y = d e_{j^*}$. Then $\sum_j s_j \chi_{a_j}(z) = d \chi_{a_{j^*}}(z) \in \{+d, -d\}$, so the spectrum of W is **exactly two points** $\{e^{\mp i\Delta d/2}\}$ at circle distance $\langle \Delta d \rangle = \gamma$. Lemma 1.3's equality case gives $\inf_{\varphi} = 2 \sin(\gamma/4)$. $\blacksquare$

Corollaries 1.8/1.9: both regimes give $\gamma = \Delta$

推论 1.8/1.9: 两种算术区间都给出 $\gamma = \Delta$

Corollary 1.8

- ① **no-wrap regime**: if $(K - 1)\Delta \leq \pi$ then for all $1 \leq d \leq K - 1$, $d\Delta \in [\Delta, \pi]$, so $\langle\langle d\Delta \rangle\rangle = d\Delta \geq \Delta$ and $\gamma = \Delta$.
- ② **cyclic regime**: if $\Delta = 2\pi/Q$ and $K \leq Q - 1$ then $\langle\langle d\Delta \rangle\rangle = (2\pi/Q) \min(d, Q - d) \geq 2\pi/Q = \Delta$, so $\gamma = \Delta$.

Corollary 1.9 (uniform separation)

In both regimes

$$\min_{x \neq y} d_{\text{proj}}(U_x, U_y) = 2 \sin(\Delta/4),$$

i.e. the K^m targets form a **packing** of separation $2 \sin(\Delta/4)$.

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Corollary 1.9 (uniform separation)

In both regimes

$$\min_{x \neq y} d_{\text{proj}}(U_x, U_y) = 2 \sin(\Delta/4),$$

i.e. the K^m targets form a **packing** of separation $2 \sin(\Delta/4)$.

In the cyclic regime the coefficient interval $[0, (Q - 2) \cdot 2\pi/Q]$ **may cross π** . A textbook principal-value argument on $[-\pi, \pi]$ would go wrong here. Theorem 1.7 handles wraparound exactly via $\langle\langle \cdot \rangle\rangle$, needing no principal-interval hypothesis. **This is the entire meaning of “exact wrap margin.”**

cyclic 情形下系数区间可能越过 π , 教科书式的主值论证在这里会出错。Theorem 1.7 用 $\langle\langle \cdot \rangle\rangle$ 精确处理绕转, 不需要主值假设——这就是“精确 wrap margin”的全部含义。

Lemma 1.10: from packing to a decoder (the pivot)

引理 1.10: 从 packing 到译码器 (枢纽)

Lemma (nearest-packing decoder)

If $\mathcal{F} = \{U_x\}_{x \in \mathcal{X}}$ is pairwise $d_{\text{proj}} \geq 4\varepsilon$ and $\hat{x}(C) := \arg \min_y d_{\text{proj}}(U(C), U_y)$ (fixed tie rule), then

$$d_{\text{proj}}(U(C), U_x) \leq \varepsilon \implies \hat{x}(C) = x.$$

若族两两分离 $\geq 4\varepsilon$, 则最近邻 packing 译码器能从任何 ε -正确的输出唯一恢复 x 。

Proof. For $y \neq x$: $d_{\text{proj}}(U(C), U_y) \geq d_{\text{proj}}(U_x, U_y) - d_{\text{proj}}(U_x, U(C)) \geq 4\varepsilon - \varepsilon = 3\varepsilon > \varepsilon \geq d_{\text{proj}}(U(C), U_x)$. ■

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► The pivot of the entire paper

It converts the soft statement “*the compiler’s output is correct*” into the hard statement “*the output uniquely determines x* ”. Hence **any correct compiler is forced to encode all of x somewhere in its output** — and information can be counted in bits. Part V does nothing but apply this conversion three times.

这条引理是全文的枢纽：把“编译器输出正确”这句软话，变成“输出唯一确定 x ”这句硬话。于是任何正确的编译器都被迫在输出里编码整个 x ，而信息可以用比特来数。

Terminology and toolkit 术语与工具字典

Packing: why distinguishable settings are countable 为什么“可区分的设置”只有可数个

► **Calibration: K_ϵ was not chosen by me** 字母表大小 K_ϵ 不是我挑的

Dispersal: no record leaks the aggregate 没有任何一条记录泄露 aggregate

The lower bounds: one argument, three variations 下界：一个论证的三次变奏

Upper bounds: a frontier, not a vacuous floor 上界：这是前沿，不是空洞的下界

The question, stated as an objection

把质疑先说出来

“Isn’t K chosen to make the bound look good?”

No: K_ε equals the metric-entropy count of pairwise distinguishable rotations at tolerance ε , up to an explicitly interpretable factor.

Definitions and the exact single-axis distance

定义与单轴精确距离

Definition

$g_\varepsilon := 4 \arcsin(2\varepsilon)$, and

$$N(\varepsilon) := \max \{ |S| : S \subseteq \{R_P(\theta)\}_{\theta \in \mathbb{R}/2\pi\mathbb{Z}}, \text{ pairwise } d_{\text{proj}} \geq 4\varepsilon \}.$$

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Lemma

$d_{\text{proj}}(R_P(\theta), R_P(\theta')) = 2 \sin(g/4)$ where $g := d_\circ(\theta, \theta')$.

Proof. $R_P(\theta)R_P(\theta')^\dagger = R_P(\delta)$, $\delta = \theta - \theta'$, whose spectrum is exactly the two points $e^{\mp i\delta/2}$ at circle distance $\langle\langle \delta \rangle\rangle = g$; apply the equality case of Lemma 1.3. ■

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Corollary 2.3

pairwise $d_{\text{proj}} \geq 4\varepsilon \iff$ pairwise $d_\circ \geq g_\varepsilon$, because $2 \sin(g_\varepsilon/4) = 2 \sin(\arcsin 2\varepsilon) = 4\varepsilon$ **exactly** — that is how g_ε was defined.

Theorem 2.4 and the cyclic exact identity

定理 2.4 与 cyclic 情形的精确恒等

Theorem (metric-entropy count)

$$N(\varepsilon) = \left\lfloor \frac{2\pi}{g_\varepsilon} \right\rfloor, \quad \frac{1}{2\varepsilon} - 1 < N(\varepsilon) \leq \frac{\pi}{4\varepsilon}.$$

容差 ε 下两两可区分的旋转个数恰为 $\lfloor 2\pi/g_\varepsilon \rfloor$, 夹在 $1/2\varepsilon - 1$ 与 $\pi/4\varepsilon$ 之间。

Proof. On a circle of circumference 2π , points pairwise $\geq g$ apart number at most $\lfloor 2\pi/g \rfloor$ (order them; the gaps sum to 2π and each is $\geq g$), attained by equal spacing. The two-sided estimate uses $x \leq \arcsin x \leq \frac{\pi}{2}x$ on $[0, 1]$:
 $g_\varepsilon \in [8\varepsilon, 4\pi\varepsilon]$. ■

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Corollary 2.5 (cyclic: exact equality)

The paper takes $Q_\varepsilon := \lfloor \frac{\pi}{2 \arcsin 2\varepsilon} \rfloor$, and

$$\frac{\pi}{2 \arcsin 2\varepsilon} = \frac{2\pi}{4 \arcsin 2\varepsilon} = \frac{2\pi}{g_\varepsilon} \Rightarrow \boxed{Q_\varepsilon = N(\varepsilon)}.$$

The sharing modulus equals the metric-entropy count itself — no free parameter at all.

sharing modulus 恰好等于 metric-entropy 计数本身——一个自由参数都没有。

Proposition 2.6: what the factor 4 is

命题 2.6: 那个因子 4 到底是什么

Proposition

With $L_\varepsilon = \lfloor \frac{\pi}{8 \arcsin 2\varepsilon} \rfloor$, $K_\varepsilon = L_\varepsilon + 1$, $\Delta_\varepsilon = \frac{\pi}{2L_\varepsilon}$:

$$(K_\varepsilon - 1)\Delta_\varepsilon = L_\varepsilon \cdot \frac{\pi}{2L_\varepsilon} = \frac{\pi}{2} \leq \pi \quad (\text{no-wrap holds}), \quad \frac{\Delta_\varepsilon}{4} = \frac{\pi}{8L_\varepsilon} \geq \arcsin 2\varepsilon \implies 2 \sin(\Delta_\varepsilon/4) \geq 4\varepsilon,$$

and with $y := \frac{\pi}{2 \arcsin 2\varepsilon}$ (so $N = \lfloor y \rfloor$, $K_\varepsilon = \lfloor y/4 \rfloor + 1$): $\frac{N(\varepsilon)}{4} < K_\varepsilon < \frac{N(\varepsilon)}{4} + \frac{3}{2}$.

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► Must answer

The factor 4 is **not slack in an estimate** — it is exactly the price of the no-wrap condition restricting the coefficient range to the quarter period $[0, \pi/2]$. The cyclic version has no such restriction and degenerates to the exact identity $Q_\varepsilon = N(\varepsilon)$. **Both directions are stated as two-sided bounds; no tighter constant is claimed.**

因子 4 不是估计里的余量，而正是 no-wrap 条件把系数范围限制到四分之一周期 $[0, \pi/2]$ 的代价。两个方向都写成双边界，未声称更紧的常数。

Remark 2.7: why $\log(1/\varepsilon)$ appears twice

注 2.7: $\log(1/\varepsilon)$ 为什么出现两次

Ross-Selinger guarantees that **every** grid angle costs $\Theta(\log(1/\varepsilon))$ in Clifford+ T synthesis, **uniformly over the alphabet**. Hence

$$\underbrace{m \log_2 K_\varepsilon}_{\text{information side: alphabet size}} = m \log_2 \Theta(1/\varepsilon), \quad \underbrace{3 \log_2(1/\varepsilon)}_{\text{physical side: repair cost per coordinate}}$$

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The two logarithms come from the **same grid**. This — not an analogy — is the technical basis on which “exchange rate” is a legitimate word.

两个对数来自同一个网格。这不是类比——它是”汇率”这个词在技术上成立的基础。

The exchange rate in miniature

汇率缩影

One QAOA cost layer, 3-regular MaxCut on 22 vertices: $m = 33$ edges, $\varepsilon = 10^{-3}$.

$L_\varepsilon = \lfloor \pi / (8 \arcsin 2\varepsilon) \rfloor = 196$, $K_\varepsilon = 197$.

Per edge: $\log_2 197 = 7.62$ bits. Whole layer: **252 bits $\approx$ 31 bytes.**

A compiler that **defers** carries those 31 bytes across every cut.

A compiler that **commits early** buys the same information back at
 $\approx 3 \log_2(1/\varepsilon) \approx 30$ T gates per corrected coordinate: **order 10^3 magic states.**

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► **Dispersal: no record leaks the aggregate** 没有任何一条记录泄露 aggregate

The lower bounds: one argument, three variations 下界：一个论证的三次变奏

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Definition 3.1: the r -round dispersed update stream

定义 3.1: r 轮打散更新流

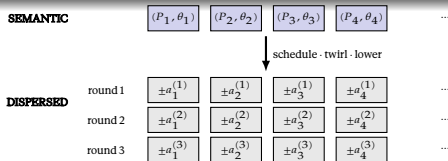
Definition (dispersed stream)

Fix $Q \geq 3, \Delta = 2\pi/Q, K = Q - 1, r \geq 2$. **Semantic input:** $x \in [K]_0^m$ — m addressed records. **Flat input:** round-major records $\text{Update}(t, j, a_j^{(t)}), a_j^{(t)} \in \mathbb{Z}_Q$, with

$$\sum_{t=1}^r a_j^{(t)} \equiv x_j \pmod{Q}, \quad j \in [m], \quad (3.1)$$

realizing $W = \prod_{t=1}^r \prod_{j=1}^m R_{P_j}(\Delta a_j^{(t)})$.

语义输入是 m 个 aggregate; 平坦输入是 round-major 的更新记录, 每个坐标在 r 轮上模 Q 求和才等于 aggregate。



no record — and no proper prefix — determines the edge total $x_j = \sum_t s_j^{(t)} a_j^{(t)} \bmod Q$

Correctness: Claim 3.2 + Proposition 3.3

正确性: 命题 3.2 与 3.3

Claim

$$R_P(\theta + 2\pi k) = (-1)^k R_P(\theta).$$

Proof. $\exp(-i\pi P)$ acts as $e^{\mp i\pi} = -1$ on both ± 1 eigenspaces of P , so $\exp(-i\pi P) = -I$; iterate. ■

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Proof. $\exp(-i\pi P)$ acts as $e^{\mp i\pi} = -1$ on both ± 1 eigenspaces of P , so $\exp(-i\pi P) = -I$; iterate. ■

Proposition

Any flat stream satisfying (3.1) realizes U_x up to a global phase.

Proof. The P_j commute, so $W = \prod_j R_{P_j}(\Delta \sum_t a_j^{(t)})$. With integer representatives $\sum_t a_j^{(t)} = x_j + Qk_j$ and $Q\Delta = 2\pi$, Claim 3.2 gives $R_{P_j}(\Delta x_j + 2\pi k_j) = (-1)^{k_j} R_{P_j}(\Delta x_j)$; the product of signs $(-1)^{\sum_j k_j}$ is global, and d_{proj} has already quotiented it. ■

Proposition 3.4: record and prefix privacy

命题 3.4: 单记录与真前缀的隐私性

Proposition

- ① For any fixed (t, j, a) and any $b \in [K]_0$, there is a valid stream with $a_j^{(t)} = a$ **and** $x_j = b$.
- ② Under the **canonical sharing distribution** (first $r - 1$ shares i.i.d. uniform on $\mathbb{Z}_Q$, last fixed by (3.1)), every individual residue is uniform on $\mathbb{Z}_Q$ **and independent of x_j** .
- ③ After any proper prefix, the final-round residue choice reaches every aggregate in $\mathbb{Z}_Q^m$, hence every $x \in [K]_0^m$.

Proof. (1) Since $r \geq 2$, pick $t' \neq t$ and set $a_j^{(t')} := b - a - \sum_{t'' \neq t, t'} a_j^{(t'')} \bmod Q$. (2) For $r = 2$: $a^{(1)}$ uniform; conditioning on x_j leaves it uniform; $a^{(2)} = x_j - a^{(1)}$ is then uniform. A sum containing at least one independent uniform term is uniform. (3) Apply (1) coordinatewise. ■

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(3.1) together with 3.4(2) **is additive secret sharing over $\mathbb{Z}_Q$** — a one-time pad. Constructing a secret sharing is cheap and is not the contribution. **The contribution is the next slide.**

(3.1) 加上 3.4(2) 就是 $\mathbb{Z}_Q$ 上的加法秘密分享，一次性密码本。构造秘密分享很便宜，不是贡献；贡献在下一帧。

Proposition 4: the FT stack emits these streams anyway

命题 4: 容错栈本来就在发这种流

Each of the following emits streams of Definition 3.1, and each generated family satisfies the hypotheses of the lower bounds:

- **(a) Time-dependent Trotterization.** $H(t) = \sum_j c_j(t)P_j$ with step τ ; round t applies $R_{P_j}(2c_j(t_t)\tau)$; grid rounding gives $a_j^{(t)}, s \equiv +1$.
- **(b) Layerwise-parameterized phase layers.** A depth- r ansatz whose round- t commuting layer carries free coefficients $\theta_j^{(t)}$.
- **(c) Randomized compiling of (a) or (b).** Twirling multiplies each residue by $s_j^{(t)}$, in folded or tracked mode.

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► The counterexample — state it yourself

The constant schedule $a_j^{(t)} \equiv a_j$ is degenerate: $x_j = ra_j \bmod Q$ is determined by round one alone. This is exactly why controlled-power ladders and phase estimation are OUT of scope: they supply contiguity but **not** coefficient freedom. An exhaustive enumeration at $(m, r, Q) = (2, 2, 5)$ confirms all three behaviours (frozen artifact `fooling_set_m2_r2_Q5.json`).

常数 schedule 是退化的——第一轮就把 aggregate 定死了。所以 controlled-power ladder 和 phase estimation 明确不在范围内：它们提供 contiguity，但不提供跨轮次的系数自由度。

Lemma 3.5: Clifford records normalize away

引理 3.5: Clifford 记录可以归一化掉

Lemma (Clifford interleaving)

Suppose the rounds are separated by Clifford records and the accumulated Clifford D_{t-1} satisfies

$$D_{t-1}^\dagger P_j D_{t-1} = \varepsilon_j^{(t)} P_{\pi_t(j)}, \quad \varepsilon_j^{(t)} \in \{\pm 1\}, \quad \pi_t \in S_m. \quad (3.2)$$

Then commuting every Clifford record to the end yields a stream of Definition 3.1 on the same generators (residues unchanged, signs multiplied by $\varepsilon_j^{(t)}$, coordinates relabelled by π_t), plus a fixed Clifford suffix D_r .

Proof. Write the emitted unitary as $D'_r L_r \cdots D'_1 L_1$. Moving each Clifford leftwards replaces L_t by $D_{t-1}^\dagger L_t D_{t-1}$; since conjugation is an automorphism and $L_t = \prod_j R_{P_j}(\Delta s_j^{(t)} a_j^{(t)})$, (3.2) gives $D_{t-1}^\dagger L_t D_{t-1} = \prod_j R_{P_{\pi_t(j)}}(\Delta \varepsilon_j^{(t)} s_j^{(t)} a_j^{(t)})$. D_{t-1} — hence $\pi_t, \varepsilon^{(t)}$ — is computable from the prefix, so the normalization is **prefix-measurable**; D_r is common to the family and unitary, preserving projective distance and the separation of Theorem 1.7. ■

Corollary 3.6: nothing depends on Pauli frames

推论 3.6: 不依赖 Pauli frame

Corollary 3.6

$a \mapsto sa \bmod Q$ ($s = \pm 1$) is a bijection of $\mathbb{Z}_Q \Rightarrow$ the twirled family maps **record-for-record** onto the untwirled family, preserving aggregates, record boundaries and cut positions.

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► Nothing depends on Pauli frames — and the boundary

Sign normalization is a **bijection of the residue alphabet**, so the standard folded (compile-time) presentation of randomized compiling inherits every bound verbatim. **Boundary (state it before being asked):** (3.2) is a real restriction. It holds for Pauli-frame layers ($\pi_t = \text{id}$) and for routing/SWAP layers permuting the generating set. **It fails** if some accumulated Clifford sends a P_j to a *product* of generators rather than a generator — the rounds then accumulate in different bases and the coordinatewise structure is lost.

符号归一化是残数字母表上的双射，folded 写法逐字继承所有界。边界要主动说：若某个累积 Clifford 把 P_j 送到生成元的乘积而不是生成元，各轮就在不同基底上累加，坐标结构随之丢失。

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Packing: why distinguishable settings are countable 为什么“可区分的设置”只有可数个

Calibration: K_ϵ was not chosen by me 字母表大小 K_ϵ 不是我挑的

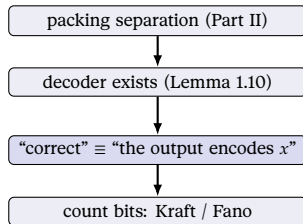
Dispersal: no record leaks the aggregate 没有任何一条记录泄露 aggregate

► **The lower bounds: one argument, three variations** 下界：一个论证的三次变奏

Upper bounds: a frontier, not a vacuous floor 上界：这是前沿，不是空洞的下界

Roadmap: one argument, three variations

路线图：只有一个论证



Variation	Setting	Tool
Thm 4.2	deterministic, one pass, zero error	fooling set + Kraft
Thm 4.4	two-round masked share, binary aggregate	same, new common-suffix construction
Thm 4.6	randomized, p passes	one-time pad + conditional Fano

Understand the argument **once**. The other two change one step of the construction and one information-theoretic fact — nothing else.

真正要理解的只有一次，后两个变奏只换一步构造、换一个信息论事实。

Definition 4.1: the cut budget and forward-pass accounting

定义 4.1: 切口预算与前向扫描计数

At every execution cut, the restart snapshot is serialized in **five disjoint fields**:

$$B_{\text{cut}} = B_{\text{control}} + B_{\text{store}} + B_{\text{window}} + B_{\text{parameter}} + B_{\text{IR}}$$

field	contents
control	finite control, pass/head/termination state, random seed + cursor
store	RAM, stacks, hash tables, caches and their layout
window	structural fields of the ordered live read window
parameter	every live parameter payload (sparse symbolic form)
IR	every other live IR/DAG/table object, and every rereadable consumed input block

Partition the round-major stream into Alice's prefix A (rounds $1, \dots, r-1$) and Bob's suffix B (round r). A forward p -pass compiler makes at most p crossings $A \rightarrow B$ and $p-1$ rewinds $B \rightarrow A$:

$$\Sigma_p := \sum_{\text{crossings}} B_{\text{cut}} \leq (2p-1)S, \quad S := \max_c B_{\text{cut}}(c). \quad (4.1)$$

► “What is $2p-1$?”

p forward crossings plus at most $p-1$ backward crossings of that one cut. That is all. **But failing to answer instantly looks like not having read your own definition.**

p 次前向穿越, 加上对同一个切口至多 $p-1$ 次回退, 仅此而已。但答不上来会显得没读过自己的定义。

Lemma 4.1: the restart lemma (where the model's power lives)

引理 4.1: 重启引理 (模型力量所在)

Lemma (restart)

*At the cut, the five-field serialization together with the already-committed output prefix **completely determines** the continuation of the run for any supplied suffix v . (For randomized runs, conditionally on the serialized random tape and cursor.)*

切口处的五字段序列化，加上已提交的输出前缀，完全决定了对任意后缀的后续运行。

Proof. `control` restores transitions, heads, dynamic pass schedule, seed cursor; `store/window` restore RAM, stacks, hash and cache metadata, live token structures; `parameter` restores every omitted coefficient by location tag; `IR` restores all remaining live intermediate objects and retained rereadable input. The five fields are disjoint and exhaustive, hence reconstruct the full machine configuration; appending the suffix executes the same continuation. ■

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► There is no escape channel

Materializing round one into a **phase-polynomial table**, a **ZX graph**, or a **cache** is **not an escape channel** — it is a **billed line item** (B_{store} or B_{IR}).

Companion guarantee (**Lemma 10, dimension-preserving instrumentation**): a reported field is exactly $8 \cdot \text{stat}(A).st_size$ bits. **There is no gate-count-to-bit code path anywhere.**

把第一轮物化成 phase-polynomial 表、ZX 图或 cache，都不是逃逸通道，而是被计费的项。全代码没有任何把门数换算成比特的路径。

Variation 1: Theorem 4.2 (zero error, one pass)

变奏一：定理 4.2（零误差、单趟）

Theorem

A deterministic one-pass compiler correct to $\varepsilon < \sin(\frac{\pi}{2Q})$ on every valid stream of Definition 3.1 satisfies

$$A_1 + B_{\text{cut}}(c_{A \rightarrow B, 1}) \geq \lceil m \log_2 Q \rceil.$$

确定性单趟编译器若在每条合法流上都 ε -正确，则已提交输出加切口快照至少 $\lceil m \log_2 Q \rceil$ 比特。

Proof. For each $u \in \mathbb{Z}_Q^m$ take the canonical prefix (put u in round one, zeros elsewhere) and set $M(u) :=$ (committed output prefix at the cut, five-field snapshot).

Claim (injectivity): $u \neq u' \Rightarrow M(u) \neq M(u')$.

Construct a common suffix. Coordinatewise, choose $v_j \in \mathbb{Z}_Q$ so that **neither** $u_j + v_j$ **nor** $u'_j + v_j$ equals the excluded residue $Q - 1$. At most two values of v_j are forbidden (namely $Q - 1 - u_j$ and $Q - 1 - u'_j$), and $Q \geq 3$, so a legal v_j exists. Then $x := u + v$, $x' := u' + v$ both lie in $[K]_0^m$ with $K = Q - 1$, and $u_j \neq u'_j \Rightarrow x_j \neq x'_j$.

Derive the contradiction. Suppose $M(u) = M(u')$. By Lemma 4.1 the continuation is a deterministic function of (snapshot, suffix), and both runs receive the same suffix v , so both emit the **same** final string C . Correctness gives $d_{\text{proj}}(U(C), U_x) \leq \varepsilon$ and $d_{\text{proj}}(U(C), U_{x'}) \leq \varepsilon$, hence $d_{\text{proj}}(U_x, U_{x'}) \leq 2\varepsilon$. But $x \neq x'$, so by Theorem 1.7 and Corollary 1.8(2), $d_{\text{proj}}(U_x, U_{x'}) \geq 2 \sin(\Delta/4) = 2 \sin(\frac{\pi}{2Q}) > 2\varepsilon$. Contradiction.

Thus M is injective on Q^m values; the messages are prefix-free under the declared serializers, so Fact K gives $\max_u |M(u)| \geq m \log_2 Q$, and $|M(u)| \leq A_1 + B_{\text{cut}}$. ■

The one clever step: why $K = Q - 1$, why $Q \geq 3$

唯一需要动脑的一步：为什么 $K = Q - 1$ 、为什么 $Q \geq 3$

► The fooling set, not the packing

$K = Q - 1$ (rather than $K = Q$) is for the fooling set, not for the packing. By Corollary 1.8(2) the separation holds just as well at $K = Q$. Excluding one residue class is what guarantees that for **any** u, u' there exists a **common** suffix v sending both into the legal alphabet.

And that is why $Q \geq 3$: at most two values of v_j are forbidden, so a third must be available.

$K = Q - 1$ 是为了 fooling set，不是为了 packing。排除一个剩余类，才能保证对任意 u, u' 都存在把两者同时送进合法字母表的公共后缀 v ；被禁的 v_j 至多两个，所以需要 $Q \geq 3$ 。

Variation 2: Theorem 4.4 (two-round masked share)

变奏二：定理 4.4（两轮掩码分享）

Setting: $Q = 3$ (sharing modulus), $\alpha = 2\pi/3$, but the **aggregate alphabet is restricted to binary**, $x \in \{0, 1\}^m$. First share $u \in \mathbb{Z}_3^m$ arbitrary, second v with $u + v \equiv x \pmod{3}$.

Theorem

A deterministic compiler correct to $\varepsilon < \varepsilon_0 := \sin(\pi/6) = 1/2$ on every valid two-round decomposition satisfies, under the single family-relative output code,

$$B_{\text{pre}}^{\text{rel}} + B_{\text{cut}}^{\text{MS}} \geq \lceil m \log_2 3 \rceil. \quad (4.2)$$

Hence if the final output is **compact**, $B_{\text{rel}}^{\text{out}} \leq m + c \log_2(m + 2)$, then

$$B_{\text{cut}}^{\text{MS}} \geq \lceil m \log_2 3 \rceil - m - c \log_2(m + 2) = \Omega(m). \quad (4.3)$$

By contrast a one-pass compiler receiving the semantic aggregate x attains $B_{\text{rel}}^{\text{out}} = m + O(\log m)$ **and** $B_{\text{cut}} = O(\log m)$.

Proof (only the step that differs from 4.2 — the coordinatewise common suffix).

$$u_j = u'_j \Rightarrow v_j := -u_j \text{ (both aggregates 0);} \quad u'_j = u_j + 1 \Rightarrow v_j := -u_j \text{ (aggregates 0, 1);}$$

$$u_j = u'_j + 1 \Rightarrow v_j := -u'_j \text{ (aggregates 1, 0).}$$

Both (u, v) and (u', v) are valid **binary**-aggregate instances and their aggregate vectors differ wherever $u \neq u'$. The rest is as in 4.2, using bit-class separation $2\varepsilon_0 > 2\varepsilon$; Kraft on 3^m prefix classes gives (4.2). ■

Why Variation 2 is not a weakening: it is a separation theorem

变奏二不是削弱，而是一个分离定理

	output	crossing state
dispersed (masked-share) stream	m bits suffice for the alphabet	but the cut must carry $m \log_2 3 \approx 1.585m$ bits, by (4.3)
semantic stream	$m + O(\log m)$	$O(\log m)$

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semantic stream	$m + O(\log m)$	$O(\log m)$

► This is the cleanest statement of the thesis

On the dispersed representation, **compact output and small state are not simultaneously achievable**; on the semantic stream **both are**.

This — not merely a lower bound — is the cleanest mathematical statement of “*representation is a resource*.” **It is a separation theorem.**

在打散表示上，紧凑输出与小状态不可兼得；在语义流上两者都能做到。这是“表示即资源”最干净的数学表述——它是一个分离定理。

Variation 3: Theorem 4.6, the main theorem

变奏三：定理 4.6，主定理

Theorem (main)

Let $Q \geq 3, K = Q - 1, 0 \leq \varepsilon < \sin(\frac{\pi}{2Q})$. Let a possibly randomized compiler be correct within projective error ε with probability $\geq 1 - \delta$ for every $x \in [K]_0^m$ and every valid r -round dispersed representation, making at most $p = p(m)$ forward passes. Then

$$\overline{B}_{\text{rel}}^{\text{out}} \geq L_\delta(m, K), \quad \overline{B}_{\text{self}}^{\text{out}} \geq L_\delta(m, K), \quad (4.4)$$

$$\overline{A}_p + \overline{\Sigma}_p \geq L_\delta(m, K). \quad (4.5)$$

可随机、 p 趟的编译器，其已提交输出 $\overline{A}_p$ 与 $(2p - 1)$ 倍切口状态 S 之和，下界是 $L_\delta(m, K)$ 。

$$\overline{A}_p + (2p - 1) S \geq (1 - \delta) m \log_2 K - h_2(\delta)$$

Proof of Theorem 4.6, part (i): the output bound

定理 4.6 证明 (一): 输出界

Take X uniform on $[K]_0^m$, fix one valid dispersed representation per value, let C be the output. By Corollary 1.9 the separation exceeds 2ε , so the nearest-packing decoder of Lemma 1.10 recovers X from a successful output with error $\leq \delta$. By Fact F',

$$I(X; C) \geq (1 - \delta)m \log_2 K - h_2(\delta) = L_\delta(m, K).$$

Also $H(C) \geq I(X; C)$, so by Fact S the expected length of either prefix-free output code is $\geq L_\delta(m, K)$; a worst-input expectation is at least the uniform average.

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This step depends only on the **output distribution** — random access, dynamic passes, and internal IR cannot weaken it.

这一步只依赖输出分布：随机访问、动态扫描次数和内部 IR 都削弱不了它。

Proof of Theorem 4.6, part (ii): the transcript bound

定理 4.6 证明 (二): 通信记录界

Put U uniform on $\mathbb{Z}_Q^m$ in round one, zeros in other prefix rounds, and set the last share

$V := X - U \bmod Q$. Given $X = x$, $V = x - U$ is uniform in U , so **V is uniform and independent of X** (one-time pad), whence $H(X | V) = H(X) = m \log_2 K$.

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Let the transcript T contain every crossing snapshot and all output chunks written on Alice's side. By Lemma 4.1 (including random tape and dynamic control), (T, V) suffices to reproduce every Bob-side segment and the complete final output, so a decoder recovers X from (T, V) with error $\leq \delta$. Conditional Fano gives

$$H(X | T, V) \leq h_2(\delta) + \delta \log_2(K^m - 1) \implies I(X; T | V) \geq L_\delta(m, K).$$

The framed transcript is prefix-free conditional on V , so its expected literal length — which is exactly $\bar{A}_p + \bar{\Sigma}_p$ — is at least this. Apply (4.1).

关键是 one-time pad: 给定最后一轮的 share V , X 仍是满熵的。再用条件 Fano, 最后用 (4.1) 把 Σ_p 换成 $(2p - 1)S$ 。



The whole of Part V in one line

整个 Part V 一句话

packing separation $\Rightarrow$ decoder exists $\Rightarrow$ “correct” $\equiv$ “the output encodes x ” $\Rightarrow$ count bits
packing 分离 $\Rightarrow$ 译码器存在 $\Rightarrow$ ”正确”等价于”输出编码了 x ” $\Rightarrow$ 数比特。

And **dispersal** guarantees that at the cut, x is not yet determined by any single record or proper prefix. So those bits **must** cross the cut: either in committed output (A_p), or in carried state (S).

Two currencies. There is no third.

dispersal 保证在切口处 x 还没有被任何单条记录或真前缀确定，所以这些比特必须过河：要么在已提交输出 A_p 里，要么在携带状态 S 里。

Remark 4.7: what is NOT claimed (say it before being asked)

注 4.7: 没有声称什么 (先说出来)

- (4.4) is an **output-information** bound; (4.5) is a **transcript** bound. **They are not interchangeable**, and neither converts self-contained circuit bits into gate counts.
- At $p = 1$ with zero error, Theorem 4.2 strengthens $m \log_2 K$ to $m \log_2 (K + 1) = m \log_2 Q$. **No such strengthening is claimed** for randomized or general multipass protocols.
- **Reverse scans, an uncharged random-access input oracle, or a different output code would be a different model.**

主动交代不声称的三件事: (4.4) 与 (4.5) 不可互换; $p = 1$ 零误差下的加强不推广到随机或多趟协议; 反向扫描、不计费的随机访问 oracle、不同的输出编码, 都属于另一个模型。

Theorem 4.8: the commitment toll (the two currencies are not at par)

定理 4.8: 承诺过路费 (两种货币不等价)

Theorem (commitment toll)

Fix $Q, K = Q - 1, \varepsilon < \sin(\frac{\pi}{2Q})$, a dispersal mask $T \subseteq [m]$ with $|T| = k$. If the compiler is correct on every valid stream and its committed prefix must be **independently parseable** (append-only first-exposure regime: an emitted record is executed and cannot be reinterpreted once the suffix arrives), then at the inter-round cut

$$A_{\text{pre}} \geq \underbrace{k \log_2(2Q)}_{\text{payload}} + \underbrace{\log_2 \binom{m}{k}}_{\text{toll}}, \quad (4.6)$$

while a deferred compiler attains $A_{\text{pre}} = 0$ with $S \leq k \log_2(2Q) + O(\log m)$. The toll is **the entropy of the commitment schedule**; it vanishes iff that schedule is determined by public data (in particular at $k = m$).

在 append-only 首次曝光机制下，提前承诺 k 个坐标要付 payload $k \log_2(2Q)$ 再加 toll $\log_2 \binom{m}{k}$ ；延迟的编译器则 $A_{\text{pre}} = 0$ 。

Proof. Two valid streams differing in round-one content (in the mask, in a residue, or in a sign) require different round-one rotations; by Theorem 1.7 their completions are separated by $2 \sin(\frac{\pi}{2Q}) > 2\varepsilon$, so a prefix executed as emitted must differ. Hence

$(T, \{a_j^{(1)}\}_{j \in T}, \{s_j^{(1)}\}_{j \in T}) \mapsto$ committed prefix is injective on $\binom{m}{k} Q^k 2^k$ values, and Kraft gives (4.6). For the deferred compiler: the round-major order of Definition 3.1 is **public**, so stored residues are addressed by position and the mask is disclosed by the round-two records **at no charge**; the $O(\log m)$ term is the header. ■

Corollary 4.9: the toll does not grow with m

推论 4.9: 过路费不随 m 增长

With $k = \eta m$, since $\frac{1}{m} \log_2 \binom{m}{\eta m} \rightarrow h_2(\eta)$:

$$\frac{1}{k} \log_2 \binom{m}{k} \rightarrow \frac{h_2(\eta)}{\eta},$$

a **constant** per generator: ≈ 2 bits at $\eta = \frac{1}{2}$, and 0 at $\eta = 1$.

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a **constant** per generator: ≈ 2 bits at $\eta = \frac{1}{2}$, and 0 at $\eta = 1$.

► The asymmetry, and its cause

Commitment is irreversible, so **you must pay the entropy of your own decision about when to commit**. A compiler that defers is handed that same decision **for free** by the stream that follows.

Three readable consequences, all direct from (4.6): **toll = schedule entropy; zero iff public; independent of m** . All three are measured (see backup).

承诺不可逆，所以你必须为“何时承诺”这个自己的决定付出熵。而延迟的编译器，是由随后的流免费把同一个决定递给它。

Terminology and toolkit 术语与工具字典

Packing: why distinguishable settings are countable 为什么“可区分的设置”只有可数个

Calibration: K_ϵ was not chosen by me 字母表大小 K_ϵ 不是我挑的

Dispersal: no record leaks the aggregate 没有任何一条记录泄露 aggregate

The lower bounds: one argument, three variations 下界：一个论证的三次变奏

► **Upper bounds: a frontier, not a vacuous floor** 上界：这是前沿，不是空洞的下界

Why this Part exists

这一 Part 为什么不可省

Without matching constructions, a lower bound is a statement about a region no one has been shown to reach.

This Part turns the bound into a frontier.

Proposition 5.1: the semantic corner

命题 5.1: 语义角

Proposition

Under the same public family dictionary, a one-pass semantic compiler receiving x packs the ordered aggregate vector into one base- K integer and streams it:

$$B_{\text{rel}}^{\text{out}} \leq \lceil m \log_2 K \rceil + O(\log(mK)), \quad S = O(\log(mK)).$$

拿到语义 aggregate 的单趟编译器，把整个向量打包成一个 K 进制整数流式输出：输出接近 $m \log_2 K$ ，携带状态只有 $O(\log(mK))$ 。这就是前沿图外那个孤立点。

Proof. Retain only a streaming index and framing state; store no coefficient table and no intermediate graph. Proposition 3.3 gives semantic equality with the flat representation. ■

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Consequence

(4.4) is tight to framing terms. Adding an input-independent abort coin (run the above with probability $1 - \delta$, else emit a fixed short circuit) gives expected relative output $(1 - \delta)\lceil m \log_2 K \rceil + O(\log(mK) + 1)$, so **the randomized output lower bound is matched up to framing and the additive $h_2(\delta)$ term.**

(4.4) 紧到 framing 项。加一枚与输入无关的 abort 硬币后，随机化的输出下界也被匹配到只差 framing 与 $h_2(\delta)$ 。

Proposition 5.2: the memory-capped block hybrid

命题 5.2: 内存受限的分块混合构造

Proposition

Fix $p, h \geq 1, q = \min\{m, ph\}$. There is a deterministic zero-error p -pass compiler aggregating q coordinates and passing the rest through in the public p -block hybrid mode, with

$$S \leq \left\lceil \frac{q}{p} \right\rceil \log_2 Q + O(\log(mQrp)), \quad A_p \leq (r-1)(m-q) \log_2 Q + O(r \log(mQrp)).$$

分块混合构造: 每趟只保留一个块的残数, 其余坐标按公开的 round-major 顺序原样透传, 所以不必保留系数表。 $h = \lceil m/p \rceil$ 时落在紧凑输出角。

At the **compact-output corner** $h = \lceil m/p \rceil$ (i.e. $q = m$):

$$B_{\text{rel}}^{\text{out}} \leq m \log_2 K + O(\log(mQrp)), \quad S \leq \left\lceil \frac{m}{p} \right\rceil \log_2 Q + O(\log(mQrp)).$$

Proof. Split the first q coordinates into at most p consecutive blocks of size $\leq h$. On pass t keep only block t 's residues and add their first $r-1$ rounds mod Q ; when the final-round mates arrive, emit the corresponding base- K aggregate block. On the first pass, emit every share of each unselected coordinate in the declared round-major order — those rotations already have the correct product (Proposition 3.3), so **no coefficient table is retained**. The live packed block, indices and framing give the bound on S . Block boundaries are a public function of (m, p, q) , so the header stores those integers rather than p separate delimiters. ■

Theorem 5.3: constant-factor tightness

定理 5.3: 常数因子紧性

Theorem

For $r = 2$ the construction satisfies $A_p + pS \leq m \log_2 Q + O(p \log(mQrp))$. Compared with (4.5) at $\delta = 0$, the ratio

$$\frac{(2p-1) \log_2(K+1)}{p \log_2 K} < 2 \log_2 3 \quad (\forall K \geq 2),$$

so *the lower envelope is attained constructively up to an absolute*

constant. $r = 2$ 时构造达到 $A_p + pS \leq m \log_2 Q + O(\cdot)$, 与 (4.5) 的比值恒小于 $2 \log_2 3 \approx 3.17$ ——下包络是被构造性地达到的。

Proof. $(2p-1)/p < 2$ and $\log_2(K+1)/\log_2 K \leq \log_2 3$ (maximal at $K = 2$). ■

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so *the lower envelope is attained constructively up to an absolute*

constant. $r = 2$ 时构造达到 $A_p + pS \leq m \log_2 Q + O(\cdot)$, 与 (4.5) 的比值恒小于 $2 \log_2 3 \approx 3.17$ ——下包络是被构造性地达到的。

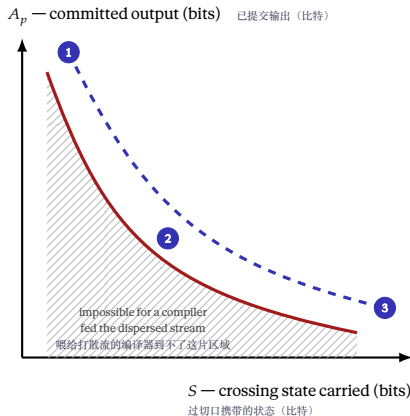
Proof. $(2p-1)/p < 2$ and $\log_2(K+1)/\log_2 K \leq \log_2 3$ (maximal at $K = 2$). ■

Absolute constant, uniform in p, m and K — so the floor is not an artefact of any particular parameter regime.

这是一个与 p 、 m 、 K 都无关的绝对常数，所以下界不是某个参数区间里的假象。

The frontier

前沿图：已证明的地板、构造达到的包络，以及不在同一输入类里的语义角



1 **materialize-first** — commits everything

全部提前承诺：输出最大，状态最小

2 **memory-capped block hybrid** — within a constant factor of the floor (Thm 5.3)

内存受限的分块混合：距地板仅一个常数因子

3 **full deferral** — $A_p = 0$

完全延迟：一个比特都不提交，全部扛在状态里

— the proved floor 已证明的地板

- - - the attained envelope 构造达到的包络

○ — **semantic input — never dispersed:** compact output, log state (Prop 5.1). *Not on the frontier: it is not in the same input class.*

语义输入，从未被打散：紧凑输出 + 对数状态。它不在前沿上，是因为它根本不属于同一个输入类——这条界不是在说编译器做不好。

Scope: the seven assumptions and where each breaks

Scope: 七条假设，以及各自在哪里失效

assumption	what breaks without it
$\mathbb{F}_2$ -independence	Lemma 1.4 surjectivity fails $\Rightarrow$ first step of Thm 1.7 cannot be executed
contiguity	commutation premise (Frame 0.4) fails $\Rightarrow$ the aggregate does not exist $\Rightarrow$ the <i>question</i> fails, not the bound
coefficient freedom across rounds	fooling set collapses to one class (constant schedule, Frame 3.4)
$Q \geq 3$	no common suffix v_j exists in Thm 4.2
$\varepsilon < \sin(\pi/2Q)$	decoder of Lemma 1.10 fails $\Rightarrow$ Fano unusable
$\delta < 1/2$	$L_\delta(m, K)$ in (0.2) loses meaning
Lemma 3.5 condition (3.2)	rounds accumulate in different bases $\Rightarrow$ coordinate structure of Def 3.1 lost

七条假设失效时坏掉的东西各不相同。特别注意第二行：失去 contiguity 时，坏掉的是“aggregate 存在”这个前提，也就是问题本身，而不是界。

Takeaway

收束

Randomize and lower late.

The information a compiler discards before aggregating cannot be recovered for free — it must be bought back **in memory** or **in magic states**, and buying it back early is strictly the more expensive of the two.

Treat recoverability as a controlled variable in compilation and resource estimation.

编译器在聚合之前丢掉的信息，不可能免费取回——只能用内存或魔术态买回来，而且早买严格更贵。请把 **recoverability** 当作编译与资源估计中的一个受控变量。

Thanks for listening.

感谢聆听

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Questions

Randomize and lower late — and treat recoverability as a controlled variable.

晚随机化、晚 lowering——并且把 recoverability 当作一个受控变量。备用帧在后面：证据分级、三个可证伪实验、scope、可复现性。

A.1 Evidence classes (Table IV)

A.1 证据分级（表 IV）

class	contents	authorized inference
① instrumented theorem-model implementation	traced reference compilers + restart verifier (592 runs)	directly tests the serialized commitment-cut coordinates; can falsify the theorems
② configured-pipeline diagnostic	matched matrix, external semantic base-lines, exact-semantics witnesses, natural workloads	compares observed tool behaviour; never proves a lower bound
③ model-specific consequence	fixed-total-error synthesis + surface-code calculation	reports consequences under the disclosed QRE; not a hardware-independent forecast

三类证据的授权推断范围不同：只有类别①能证伪定理；类别②只比较工具的实际行为，永远不能证明下界；类别③只在已披露的 QRE 假设下报告后果，不是硬件无关的预测。

A.2 Only three experiments can falsify the theory

A.2 只有三个实验能证伪理论

Fig 5 — trace-derived frontier (592 runs) → tests Theorem 4.6. Measures the theorem's own coordinates A_p , S , IR share; the black edge is the measured Pareto set, the theoretical lower envelope evaluated **in the same serialized-bit coordinates**. All frozen points lie in the theorem-allowed region; no point removed as an outlier. *Falsification: any point below the envelope kills Theorem 4.6.* (Echo occupies the early-commit corner; full aggregation the state-heavy compact-output corner; three capped hybrids interpolate — matching the three constructions of Part VI.)

Fig 2 — η sweep → tests Proposition 6 and “two currencies”. Left: a deferred compact-output compiler pays dispersal in crossing state; slopes 10.3/21.1/43.3 bytes per unit η at $m = 32/64/128$, sitting on the zero-error floor $\eta m (\log_2 3 + 1)/8$ bytes (ratios 0.99–1.05). *Read the floor:* $\log_2 3 = 1.585$ bits of residue + 1 bit of sign/frame = 2.585. Right: a streaming compiler pays the same dispersal in committed prefix (slopes 60/129/260, larger by the public addresses) while its crossing state stays flat at ≤ 8 bytes. *Falsification: no tradeoff between panels, or state channel below the floor.*

Fig 3 — commitment toll, preregistered → tests Theorem 4.8 + Corollary 4.9. Four serializers at $m \in \{128, 512, 2048\}$ (two up to $m = 32768$), every cell restart-verified: matches $\log_2 \binom{m}{k}/k$ within 0.17 bits worst case, 0.02 typically; **flat in m** at fixed η (2.11/1.99/1.97 at $\eta = \frac{1}{2}$ across a $16\times$ range); **exactly zero at $\eta = 1$** ; **exactly zero at every η in the control** (prefix omits the mask, recovers it from the suffix). *Falsification: toll growing with m , or nonzero in the control, or nonzero at $\eta = 1$.*

三个实验，三条证伪条件：Fig 5 只要有一点落在包络之下，Theorem 4.6 就死；Fig 2 只要两个面板之间没有 tradeoff、或状态通道低于地板；Fig 3 只要 toll 随 m 增长、或对照组非零、或 $\eta = 1$ 时非零。

A.2 (cont.) Fig 3: what the control experiment shows

A.2 (续) Fig 3: 对照实验说明了什么

The control (prefix omits the mask, recovers it from the suffix) is **exactly zero at every η** . This shows that **irreversibility of commitment, not addressing, is what buys the asymmetry**.

对照组（前缀不写 mask，由后缀恢复）在每个 η 上都恰好为零。这说明买到不对称的是承诺的不可逆性，不是寻址。

Right panel: under an identity-optimal serializer the ratio is 1.00 at $\eta = 1$ and ≤ 2.64 throughout — the 6:1 of Fig 2 is **codec slack**, and only the slack grows with address width (5.4:1 at $m = 128$ to 10.8:1 at $m = 32768$).

The preregistration recorded “there is no address-width law here” as a prediction against the natural reading of Fig 2, and it is reported as such.

A.3 Diagnostics (class ②): what they do and do not show

A.3 诊断实验 (类别 ②): 能说明与不能说明什么

- **Fig 8 + Table VII — mechanism ablation.** With Fourier-layer IR disabled, the compiler completes at 4k/10k/20k but materializes outputs of the same order as `qiskit_opt3`, and exceeds the 600 s budget at 50k/100k; enabled, it returns the same 42-gate circuit at every scale. **The constant form is caused by semantic lifting + coefficient aggregation, not by a lucky pass ordering.** (This is the only experiment answering that objection.)
- **Fig 6/7 + Tables XIII/XIV — exact-semantics witnesses** (Corollary 1, $\varepsilon = 0$, fixed width). Semantic UCC constant (42 rational / 27 irrational). **Rebased TKET PauliSimp recovers a small validated form at all five scales (52-73 / 35-38)** — the classification's positive prediction. `staq/qiskit_opt3` / GuidedPauliSimp grow linearly. **None of these is an impossibility result.**
- **Fig 4 — matched matrix (1080 cells).** Same target, basis, budget, seed; **only the representation varies.** (a) certified completion probability, (b) serialized output size among completed cells only. masked-share $\times$ Qiskit L3 completes at 0.38 (`predicate_error`), **retained as a status observation, never scored as a numerical loss.**
- **Tables V/VI.** Table V is explicitly **not** an instance of the masked-share lower bound (balanced encoding tokens contain x_i/r , hence leak the aggregate). Table VI tests Prop 13: output constant in r , $O(m_{CP}) = O(n(n-1)/2)$ across widths.
- **Certificate audit.** 56 adversarial mutations (angle, sign, support bit, qubit permutation, Clifford frame, deletion, duplication) with **0 false accepts**; 12 dense/symbolic cross-checks at $n \leq 6$ with no disagreement. This is what makes “only `completed_valid` enters quality means” meaningful.

这一整页都是类别 ②: 只比较工具的实际行为, 不构成任何不可能性结果。Fig 8 是唯一回答“是不是碰巧的 pass 顺序”的实验; Fig 6/7 里外部 TKET 恢复出小形式, 是分类的正面预言, 要主动承认。

A.4 The E7 preregistered TKET scan: report, do not repair

A.4 E7 预登记 TKET 扫描：只报告，不修补

$m \in \{8, 16, 32, 64, 128\}$, three seeds, byte-identical pass sequence. TKET achieved compact output in **every** cell (508 dispersed rotations $\rightarrow$ 127 non-Clifford R_z at $m = 128$); its pass-phase memory and serialized IR grew with log-log slope ≈ 1.7 in m , while the instrumented semantic-input control stayed flat (196 $\rightarrow$ 199 bytes over a $16\times$ range).

The preregistered point prediction of exactly linear growth was rejected under its own frozen decision rule, and is reported as rejected.

预登记的”严格线性增长”点预测，在它自己的冻结判定规则下被否决，并按被否决如实报告。

Two scope limits, stated explicitly:

- ① What is measured at a third-party API boundary is a pass-phase memory and serialized-IR snapshot — **class ②; not promoted to a measurement of B_{cut} , and not commensurable with the S of the frontier.** The scan tests only the qualitative dichotomy (dispersed input forces growth in m ; semantic input does not) — neither the constant nor the exponent.
- ② Superlinear growth means **the floor is not the binding constraint on this tool:** the bound establishes the floor, the observed cost exceeds it, and the excess is implementation structure (the pairwise commutation structure of a Pauli graph is itself superlinear in the number of terms). **No claim that our implementation outperforms these tools.**

A.5 Consequences (class ③): the QRE campaign

A.5 后果 (类别 ③): QRE campaign

Table VIII + Fig 9 + Fig 13, **288 rows**. One fixed $\varepsilon_{\text{total}}$ split additively: 0.10 algorithmic, 0.45 rotation synthesis, 0.45 logical failure. **Every angle actually synthesized** by a frozen deterministic GMP build of `staq grid_synth` (670 angle-precision pairs; reported error never exceeds the allocated radius).

Across all 96 matched target/error/allocation/QEC/factory settings: materialize-first uses **385-1753×** the logical T states and **538-5573×** the spacetime volume of semantic-first. **No reversal in the scanned grid.**

Two things to volunteer

- ① This is **not** a claim that our implementation dominates other semantic tools — external **TKET PauliSimp uses 0.443-0.536×** **our T states** on this witness, i.e. it is better, and crediting that recovery is part of the comparison.
- ② Absolute physical numbers still depend on routing, correlated errors, factory layout, feed-forward and code-distance assumptions. **Not a hardware-independent forecast.**

两件要主动交代：外部 TKET PauliSimp 在这个 witness 上只用我们 0.443-0.536 倍的 T 态，即它更好；绝对物理数字仍依赖 routing、相关误差、factory 布局等一整套已披露假设。

A.6 Honest nulls and anti-regression

A.6 诚实的零结果与防回归

- **Fig 10 / Table IX — real instances.** Parity with `qiskit opt3`; $\approx 65\%/73\%/78\%$ gate reduction against the upstream UCC v0.4.12 pre-semantic default. **The paper explicitly states this is an anti-regression check, not evidence of semantic recovery.**
- **Fig 11/12 + Table X — W8 six-factor ablation (21,060 cells).** Preset recognizer largest main effect (mean reduction 364.9, bootstrap CI [349.8, 380.5]); **selector, cache/reuse and projected-block selection have exactly zero isolated effect.** Both negative controls show zero all-on-minus-all-off differences in gates, depth, CX, logical T and spacetime. **Five structured families meet the predeclared all-size/all-seed criterion; QFT arithmetic and phase-polynomial blocks do not and are retained in the table.**

零结果照登不删：Fig 10 只是防回归检查，不是语义恢复的证据；W8 消融里三个因子的孤立效应恰好为零，两个族没有达到预先声明的标准，都保留在表里。

A.7 The falsifiability one-liner

A.7 可证伪性一句话

► If you remember one defensive sentence

Only three experiments can falsify my theorems: Fig 5 (the frontier envelope), Fig 2 (the two channels and the floor), Fig 3 (the four predictions of the toll).

Everything else is either a configured-pipeline diagnostic or a consequence under a disclosed model — **I use none of it to support any lower bound.**

Two third-party preregistrations failed; I report them rather than repair them.

只有三个实验能证伪我的定理：Fig 5、Fig 2、Fig 3。其余要么是 configured-pipeline 诊断，要么是已披露模型下的后果——我不用它们支持任何下界。两个第三方预登记失败了，我如实报告而不修补。

A.8 Reproducibility (only if asked)

A.8 可复现性（被问到才讲）

- Canonical registry data/manifest.yaml; normalized Parquet index with **24,748 unique records**, retaining **48 predicate-error rows outside quality means**.
- Primary key: experiment_id, instance_id, representation_id, method_id, seed, backend_hash, tool_version, artifact_commit, config_hash.
- Every printed number traced to a unique source row via data/provenance_map.yaml, tested by tests/test_paper_number_provenance.py.
- One-command entry point reproducibility/run_submission.sh: reruns audits, regenerates all tables/figures, executes the test suite, builds the manuscript, rejects unresolved references.
- Clean-room runner: independent temporary directory + fresh environment on locked packages; reconstructs 9 headline-theorem tests, one certified QAOA/Ising instance through both paths, 17 deterministic synthesis pairs. **This is working-directory isolation, not host-binary independence.**

每一个印出来的数字都能经 provenance_map.yaml 追到唯一源行，并有测试守着。最后一条要主动说：clean-room 是工作目录隔离，不是宿主二进制独立。